DIVISORS AND DIFFERENTIALS ON CURVES

In this lecture, we continue our discussion of divisors on smooth projective curves and then introduce rational differentials and their divisors. As before, we will not prove all the things covered here, since these lectures are to be taken partly as a glossary of terms and results that we will use. The students are expected to work out omitted details on their own, as this will give them much needed experience with these objects.

1. Pullback and pushforward of divisors

Let $\phi : C_1 \longrightarrow C_2$ be a nonconstant morphism of smooth projective curves over $\bar{K}$.

Definition 1.1. For a point $Q \in C_2$, define

$$\phi^*[Q] = \sum_{P \in \phi^{-1}(Q)} e_\phi(P)[P],$$

and extend $\mathbb{Z}$-linearly to obtain $\phi^* : \mathrm{Div}(C_2) \longrightarrow \mathrm{Div}(C_1)$. This is called the *pullback of divisors*.

For a point $P \in C_1$, define

$$\phi_*[P] = [\phi(P)],$$

and extend $\mathbb{Z}$-linearly to obtain $\phi_* : \mathrm{Div}(C_1) \longrightarrow \mathrm{Div}(C_2)$. This is called the *pushforward of divisors*.

Remark 1.2. The definitions above use geometric points over $\bar{K}$. For divisors written in terms of closed points over a non-algebraically-closed field, residue-field degrees must be included in the definition of pushforward.

Example 1.3. Let $f \in \bar{K}(C)$ be a nonconstant function that corresponds to a morphism $f : C \to \mathbb{P}^1$. Then it follows from the definition that $\mathrm{div}(f) = f^*(D)$, where D denotes the divisor $0 - \infty$ on $\mathbb{P}^1$.

Proposition 1.4. *Let $\phi : C_1 \to C_2$ be a nonconstant map of smooth curves. Let $D \in \mathrm{Div}(C_2)$ and $E \in \mathrm{Div}(C_1)$.*

(1) $\deg(\phi^* D) = (\deg \phi)(\deg D)$.
(2) $\deg(\phi_* E) = \deg E$.
(3) $\phi_* \phi^* D = (\deg \phi) D$.
(4) For every $f \in \bar{K}(C_2)^\times$, we have $\mathrm{div}(\phi^* f) = \phi^* \mathrm{div}(f)$.
(5) For every $g \in \bar{K}(C_1)^\times$, we have $\mathrm{div}(\phi_* g) = \phi_* \mathrm{div}(g)$.
(6) If $\psi : C_2 \to C_3$ is another nonconstant morphism of smooth curves, $(\psi \circ \phi)^* = \phi^* \circ \psi^*$ *and* $(\psi \circ \phi)_* = \phi_* \circ \psi_*$.

Proof. The first and third assertions follow from

$$\sum_{P \in \phi^{-1}(Q)} e_\phi(P) = \deg \phi.$$

The second follows directly from the definition of pushforward. For the fourth, if $P \in C_1$ and $Q = \phi(P)$, then

$$\mathrm{ord}_P(f \circ \phi) = e_\phi(P) \, \mathrm{ord}_Q(f).$$

The fifth assertion is the corresponding valuation formula for the norm map. The final assertion follows from definitions. □

Corollary 1.5. *Pullback and pushforward preserve principal divisors. Consequently, they induce homomorphisms on divisor class groups. In particular, pullback induces homomorphisms*

$$\phi^* : \mathrm{Pic}(C_2) \longrightarrow \mathrm{Pic}(C_1)$$

and

$$\phi^* : \mathrm{Pic}^0(C_2) \longrightarrow \mathrm{Pic}^0(C_1).$$

2. Rational differentials on a curve

Now we introduce the notion of vector space of differentials on a curve. This notion 'linearizes' the curve, which is extremely useful for many applications in arithmetic. Another useful application is that these help detect when a morphism is separable. This is usually read off from the derivative of a function, and naturally differentials on a curve carry out that function in this setting.

Definition 2.1. The space of *rational (or meromorphic) differentials* on C is the $\bar{K}$-vector space of Kähler differentials $\Omega_{\bar{K}(C)/\bar{K}}$. It is generated by symbols df, for $f \in \bar{K}(C)$, subject to the relations

(1) $d(f+g) = df + dg$,
(2) $d(fg) = f\,dg + g\,df$,
(3) $da = 0$ for every $a \in \bar{K}$.

We will usually denote this space simply by $\Omega(C)$.

For $\phi : C_1 \to C_2$ a nonconstant morphism of curves. Then this induces a pullback on differentials $\phi^* : \Omega(C_2) \to \Omega(C_1)$ much the same way it does so on functions:

$$\phi^*\big(\sum f_i dx_i\big) = \sum (\phi^* f_i) d(\phi^* x_i).$$

Proposition 2.2. *Let C be a smooth curve as before.*

(i) The space $\Omega(C)$ is a one-dimensional vector space over $\bar{K}(C)$.
(ii) More precisely, if $t \in \bar{K}(C)$, then $\Omega(C) = \bar{K}(C)\,dt$ if and only if $\bar{K}(C)/\bar{K}(t)$ is a finite separable extension.

Proof. The proof of the first assertion is essentially that Kahler differentials for a smooth curve are (isomorphic naturally with) (linear) dual to the tangent space at any smooth point, i.e. they form the cotangent space. In fact, this statement is local by its very nature, and so it holds at every (smooth) point P: we can define the module of Kahler differentials defined (or regular) at P as a module over $\mathcal{O}_{C,P}$ which is a DVR, and its rank over this DVR is also equal to 1.

The second assertion is essentially calculus: it says that the formal derivative of a separable polynomial cannot vanish. □

With that last comment, the next corollary should not be too surprising.

Corollary 2.3. *$\phi : C_1 \to C_2$ a morphism of smooth curves is separable if and only if $\phi^* : \Omega(C_2) \to \Omega(C_1)$ is injective.*

Remark 2.4. If $\mathrm{char}(\bar{K}) = 0$, then $df = 0 \iff f \in \bar{K}$. If $\mathrm{char}(\bar{K}) = p > 0$, then $df = 0 \iff f \in \bar{K}(C)^p$. In particular, $d(f^p) = 0$. This is the differential manifestation of the special role played by Frobenius in characteristic p.

Recalling the role uniformizers at a smooth point play, the following corollary also easily follows.

Corollary 2.5. *In particular, if t is a uniformizer at any point of C, then every rational differential can be written uniquely in the form $\omega = g\,dt$, for some $g \in \bar{K}(C)$. We denote this as $g = \omega/dt$.*

Remark 2.6. Let $f \in \bar{K}(C)$ be regular at P. Then if we set $\omega = df$, the function $\omega/dt = df/dt$ is also regular at P. This follows from the assertion made above about rank of the module of Kahler differentials at P.

Definition 2.7. Let $0 \neq \omega \in \Omega(C)$. Write $\omega = g\,dt$ with respect to a uniformizer t at P. We define

$$\mathrm{ord}_P(\omega) := \mathrm{ord}_P(g).$$

This definition is independent of the choice of the uniformizer, because if t' is another uniformizer at P, dt'/dt and dt/dt' are both regular at P by the previous remark, so their order at P must be 0. We also set $\mathrm{ord}_P(0) = \infty$. If $\mathrm{ord}_P(\omega) > 0$, we say that ω has a *zero* at P. If $\mathrm{ord}_P(\omega) < 0$, we say that ω has a *pole* at P. If $\mathrm{ord}_P(\omega) \geq 0$, we say that ω is *regular* at P.

We have the following interesting computation regarding orders.

Proposition 2.8. *Let $p =$ char K where we allow it to equal 0 if necessary. Let $P \in C$. For $x, f \in \bar{K}(C)$ with $x(P) = 0$. we have:*

(i) $\mathrm{ord}_P(fdx) = \mathrm{ord}_P(f) + \mathrm{ord}_P(x) - 1$, *if* $p = 0$ *or* p *does not divide* $\mathrm{ord}_P(x)$,
(ii) $\mathrm{ord}_P(fdx) \geq \mathrm{ord}_P(f) + \mathrm{ord}_P(x)$, *if* $p > 0$ *and* $p \mid \mathrm{ord}_P(x)$.

Proof. Write $x = ut^n$ for u a unit in the local ring and $n = \mathrm{ord}_P(x)$. Then $dx = [nut^{n-1} + (du/dt)t^n]dt$. Hence, if the first term does not vanish, then $\mathrm{ord}_P(fdx) = \mathrm{ord}_P(f) + n - 1$, which straightaway gives the first result. If $p > 0$ and $p \mid n$, then $\mathrm{ord}_P(fdx) = \mathrm{ord}_P(f(du/dt)t^n dt) \geq \mathrm{ord}_P(f) + n$. □

Proposition 2.9. *A nonzero rational differential* ω *on a smooth curve* C *has only finitely many zeroes and poles.*

Proof. Choose some t in $\bar{K}(C)$ such that $\bar{K}(C)/\bar{K}(t)$ is a finite separable extension and write $\omega = f\,dt$. The function f has only finitely many zeroes and poles. The morphism $t : C \longrightarrow \mathbb{P}^1$ also has finitely many poles and also finitely many points where it ramifies. (This statement can be seen as an analogue of the algebraic number theory statement. It can be proved in a similar manner, since this is really a fact about Dedekind domains and DVRs.) For any P away from all of the finitely many points above, in particular the element $t - t(P)$ is a uniformizer at P and hence the previous formula computes $\mathrm{ord}_P(fdx) = 0$ at all such P. □

Similar to functions, we can also define divisors of differentials. In fact, they will play a rather special role in the future.

Definition 2.10. Let $0 \neq \omega \in \Omega(C)$. The *divisor of* ω is

$$\mathrm{div}(\omega) = \sum_{P \in C} \mathrm{ord}_P(\omega)[P].$$

The differential is called *regular or holomorphic*, if $\mathrm{ord}_P(\omega) \geq 0$ for all $P \in C$, and it is *nonvanishing* if $\mathrm{ord}_P(\omega) \leq 0$ for all $P \in C$.

If ω_1, ω_2 are nonzero differentials, then there exists a function $f \in \bar{K}(C)^\times$ such that $\omega_1 = f\omega_2$, since the space of differentials is one dimensional. Then $\mathrm{div}(\omega_1) = \mathrm{div}(f) + \mathrm{div}(\omega_2)$, so that the following definition makes sense.

Definition 2.11. A divisor of the form $\mathrm{div}(\omega)$ is called a *canonical divisor*. Its divisor class in $\mathrm{Pic}(C)$ is called the *canonical divisor class* of C.